

Practical # 10

Objective:

Apply binary search algorithm and evaluate its efficiency compared to linear search.

Theory:

Binary search algorithm finds a given element in a list of elements with $O(\log n)$ time complexity where n is total number of elements in the list. The binary search algorithm can be used with only a sorted list of elements. That means the binary search is used only with a list of elements that are already arranged in an order. This search process starts comparing the search element with the middle element in the list. If both are matched, then the result is "element found". Otherwise, we check whether the search element is smaller or larger than the middle element in the list. If the search element is smaller, then we repeat the same process for the left sublist of the middle element. If the search element is larger, then we repeat the same process for the right sublist of the middle element. We repeat this process until we find the search element in the list or until we left with a sublist of only one element. And if that element also doesn't match with the search element, then the result is "Element not found in the list".

Binary search is implemented using following steps:

Step 1 - Read the search element from the user.

Step 2 - Find the middle element in the sorted list.

Step 3 - Compare the search element with the middle element in the sorted list.

Step 4 - If both are matched, then display "Given element is found!!!" and terminate the function.

Step 5 - If both are not matched, then check whether the search element is smaller or larger than the middle element.

Step 6 - If the search element is smaller than middle element, repeat steps 2, 3, 4 and 5 for the left sublist of the middle element.

Step 7 - If the search element is larger than middle element, repeat steps 2, 3, 4 and 5 for the right sublist of the middle element.

Step 8 - Repeat the same process until we find the search element in the list or until sublist contains only one element.

Step 9 - If that element also doesn't match with the search element, then display "Element is not found in the list!!!" and terminate the function.

In this lab, students will learn how to:

- Develop C++ programs to implement the Binary Search algorithm.
- Apply Binary Search to sorted arrays and strings.
- Analyze the efficiency of Binary Search in terms of time complexity (best, average, and worst cases).
- Evaluate the number of comparisons in Binary Search vs Linear Search.
- Compare the performance of Binary Search and Linear Search for datasets of varying sizes.

C++ program: Write C++ program for binary search algorithm.

```
#include<iostream>
using namespace std;
int main()
{
    int i, arr[10], num, first, last, middle;
    cout<<"Enter 10 Elements (in ascending order): ";
    for(i=0; i<10; i++)
        cin>>arr[i];
    cout<<"\nEnter Element to be Search: ";
    cin>>num;
    first = 0;
    last = 9;
    middle = (first+last)/2;
    while(first <= last)
    {
        if(arr[middle]<num)
            first = middle+1;
        else if(arr[middle]==num)
        {
            cout<<"\nThe number, "<<num<<" found at Position "<<middle+1;
            break;
        }
        else
            last = middle-1;
        middle = (first+last)/2;
    }
    if(first>last)
        cout<<"\nThe number, "<<num<<" is not found in given Array";
    cout<<endl;
    return 0;
}
```

OUTPUT

```
Enter 10 Elements (in ascending order): 12
13
14
15
16
17
18
19
20
21

Enter Element to be Search: 17

The number, 17 found at Position 6
```

Review Questions/ Exercise:

1. Write a C++ program to implement Binary Search on an integer array.
2. Modify the program to search for a string element in a sorted list of names.
3. Compare the number of iterations required in Linear Search vs Binary Search for the same dataset.
4. Develop a program that reads a sorted dataset and applies Binary Search to find an element. Print the total number of comparisons made.
5. Construct a program to test Binary Search on large input arrays.

Name: _____

Roll #: _____

Date: _____

Subject Teacher